

Problem 8.14

Suppose that $a(t) = 1.12^t$. What is the effective rate of discount in the 7th year?

Solution

The effective rate of discount in the 7th year is the amount of growth during that year divided by the accumulated value at the end of the year:

$$d_7 = \frac{a(7) - a(6)}{a(7)}.$$

Substituting $a(t) = 1.12^t$ gives

$$\begin{aligned} d_7 &= \frac{1.12^7 - 1.12^6}{1.12^7} \\ &= 1 - \frac{1}{1.12} \\ &\approx 0.107142857. \end{aligned}$$

Therefore,

$$\boxed{d_7 \approx 10.714\%}$$